LECTURE 4: EXAMINATION SYSTEM IN KENYA

Introduction

Welcome to lecture 4. We are going to discuss the current place of the national examination in our country and further look at the merits and the demerits of the examination system. The performance in most exams is normally skewed and fails to take the normal assumption as predicted in mot studies this will also be explained further.

Learning objectives

After this lecture, the learner should be able to:

a. Discuss the current place of national examinations in Kenya.
b. Discuss public criticisms of national examinations in Kenya.
c. Explain non-normality in examinations.
d. Discuss the proposed Competency Based Education.

Since the colonial era, examinations have been part and parcel of education in Kenya. During the colonial period secondary school examinations for Form IV and Form VI were conducted by Cambridge Examinations Syndicate.

Why do we need national assessments?

1. We need to know how well students are learning especially in regard to the:
   (a) National education goals; (what are our national education goals).
   (b) Aims of the curriculum.
   (c) Preparation for further learning and lifelong learning.

In 2016, only 15% achieved between A and C+. This is a group that demonstrated mastery of subject content and the readiness for further learning. The implication of 15% is that affirmative action policy will not be applied this year in university admission. If applied the C grade will be admitted.

2. The examination should act as an assessment of students' knowledge and skills. A tool for testing that a certain level of education has been achieved by the learner. This is an important aspect for diagnosing learning competencies and placement of students in the right career and/or academic tracks; identify students who need remedial teaching.
3. Assessment can help us understand inadequacies of learning outcome by subgroups such as sex, geographical region and social class. It would be of grave concern to the public and Ministry of Education if, for example, the 6% who achieved grade E and the 45% between D and D- belongs to certain geographical regions – those from ASAL poor.
4. The examination results could be used to interrogate what factors are associated with performance such as student facilities, student-teacher ratio, student-textbook ratio, school leadership, etc. The 2016 KCSE results clearly indicated that some of these factors require scrutiny.

Colonial Period

1925: Nairobi European School did poorer than Indian schools in KCE. The results were suppressed. The white community did not want to hear that Indian students have done better than the whites.

1940: Alliance High School and Mangu High School did better than Prince of Wales School. The white community was unhappy.

Post-Colonial

After independence, Kenya, Uganda and Tanzania under the umbrella of East African Community established the East African Examinations Council to take over from Cambridge Examinations Syndicate.

Due to ideological differences between the three countries, particularly between Kenya and Tanzania (capitalism of Kenya and socialism of Tanzania) East African Community was disbanded in 1977 and each country established its own examination board.

Briefly, between 1977 and 1980, examinations were conducted by the examinations Department in the Ministry of Education.

KNEC Legal Mandate

In 1980, KNEC was legally established. KNEC is legally mandated inter alia to:

(i) Conduct all school examinations.

(ii) Authorize other foreign examining boards to conduct their examinations in the country.

(iii) Issue certificates and diplomas to successful candidates.

(iv) Ensure security of examinations

(v) Take legal and disciplinary actions against those involved in examination malpractices, including prohibiting a candidate who cheats in examination from taking an examination for a period not exceeding three years. Universities should also consider this option.

(vi) The 1980 KNEC Act was repealed in 2012 and further amendments made in 2015. The 2015 Amendment created National Examinations Appeal Tribunal to deal with appeals from aggrieved parties on withdrawing, nullification and cancellation of results.

Since independence, the various education commissions that have been set to look into educational reforms have all made recommendations on the examination system. These recommendations included:

(i) Change of examination name

- Kenya African Preliminary Examinations (Colonial name)
- Kenya Primary Education (post-colonial name)
- Certificate of Primary Education (post-colonial name)
- Kenya Certificate of Education (For Form IV)
- Kenya Advanced Certificate of Education (For Form VI)
- Kenya Certificate of Secondary Education from 1984.
- Kenya Certificate of Basic Education (KCBE) for 2-6-3-3 system. To be administered in Form IV.

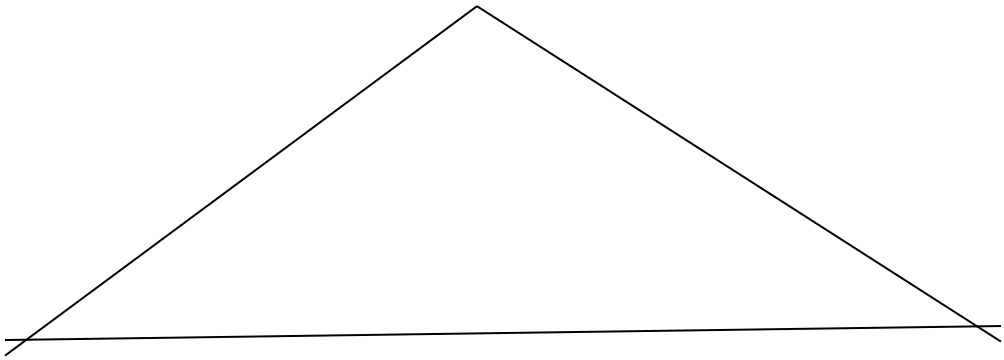
(ii) Number and combinations of subjects to be offered in the examination.

The current main examinations that are conducted by KNEC are:

1. KCPE
2. KCSE
3. Primary Teachers Examinations (PTE), Technical and Business at craft level, certificate, diploma and higher diploma.

Discuss triangular relationship between MoE (policy issues), KICD (curriculum development and review) and KNEC (examination of the achievement of the curriculum).

MoE (Policy Issues)



KICD TVET CDACC **KNEC**

(Curriculum Development) (Examining the Achievement of Curriculum)

Organizational Structure of KNEC

On policy decision, KNEC has a Council. The council members are appointed by the Cabinet Secretary in accordance with KNEC Act. Besides general administration, the main departments at KNEC are:

- Test development department.
- Test administration department.
- Research department.

The secretariat is headed by Council Secretary (CEO) who is responsible for day-to-day running of KNEC. The key officers in KNEC are subject officers/subject experts. The subject officers are people who have been teachers. The responsibilities of the subject officers include:

(i) Being in charge of specific subject, e.g. CRE, biology, English, mathematics

(ii) Participating in curriculum development at Kenya Institute of Curriculum Development

(iii) Identifying test developers.

(iv) Development of the table of specifications.

(v) Organizing moderation and Awards Sessions.

KNEC's Role in Conducting School-Examinations

With respect to examinations, KNEC has four primary roles in ensuring that the national examinations are fair to all candidates, namely:

- Examinations development
- Examination Administration
- Examination processing and issuing certificates and diplomas
- Preparation of post-mortem reports

Examination Development

Examination development is a process that starts with test planning to moderation of draft questions. The subject experts are guided through the table specifications by the subject officer. The subject expert uses the syllabi to prepare the test items. Once the test items have been prepared, they are subjected to moderation by another team of subject experts. Moderation is a process of ensuring that test items are not ambiguous and are free of errors.

Item Banking

Discuss the concept of ***Item Banking and its importance and challenges in a corrupt society like Kenya.***

- Globally used by examination boards in Europe
- KNEC used item banking until 2016 when CS ignorantly directed KNEC to abolish it.
- Used at the Open University of Tanzania
- Useful when the curriculum has stabilized
- In case of major disaster like plane crush while bringing exams from UK where they are printed and mass leakage that may lead to cancellation of the entire examination item banking is useful

Examination Administration

This is a process of field administration of the examinations to ensure that no candidates receive unfair advantage. This involves:

(i) Recruitment of examiners, supervisors and invigilators.

(ii) Briefing of supervisors and invigilators.

(iii) Distribution of question papers to all examination centres

(iv) Ensure that transport for examinations and personnel including security during examination period in place.

(v) Training of examiners on marking skills and co-ordination of the entire examination processing.

(vi) Compiling records from field reports of examination irregularities encountered.

(vii) Security of examination for the purpose of integrity of examinations.

Examination Processing

Examination processing is an exercise that involves:

(i) Marking being the main activity. KNEC examinations marking/scoring is in two parts, namely:

(a) Manual marking for non-objective test items (essay)

(b) Mechanical marking for objective test items using Optical Mark Readers (OMR)/ scanners. Also referred to as Optical Mark Recognition.

Advantages* of Optical Mark Readers (*OMR)

- Speed
- Accuracy
- Cost –efficiency - no markers

(ii) Grade awards. Determining grade levels for various subjects.

Post-mortem Reports

After the examination, KNEC carries out post facto analysis of the candidates' performance per subject. The purposes of this exercise are:

- To identify the general performance trend of the candidates.
- To identify the test items that attracted poor performance and that reflect poor teaching.
- To assist teachers, know the areas where candidates performed poorly.

What will happy to 2016 KCSE results since the government is hiding subject by subject performance?

Criticisms and Effects of Examinations

The criticisms and the effects of KNEC examinations:

- Teachers' use of examinations to teach and hence corrupting teaching.
- Promotion of teachers on the basis of good results.
- Demotion of Head teachers on poor results.
- Transfer of teachers and Head teachers on poor results
- Blocking of Head teachers' offices by parents on poor results
- That the examinations are promoting rote learning.
- Elitism. That examination favour children from privileged backgrounds.
- That the examination setters come from high performing schools.
- Makes education examinations oriented. Since independence this is the criticism that has led to education reform.
- Brings unhealthy competitions between schools whereby private schools use examination results for commercial purposes. Poor learners are directed to do examinations outside their own schools. Cheating has also been associated with pressure to get good grades.
- Whole destiny of the child's life is determined by the national examinations results
- Increase coaching and drilling of students.
- Motivation for subject teachers. Teachers are rewarded in form of provision of trips, money or when their students perform in their subjects.
- That the examinations are manipulated to favour areas in regions where powerful politicians come from.
- That ethnic bias may enter into marking through the examiners knowledge of the candidates' names or school.
- ***School ranking***: Schools are not the same in terms of teaching and learning resources and hence ranking is unrealistic.

- ***Testing primarily cognitive domain***: Skills and values and attitude are rarely tested.
- ***Seat-time education/time-based education***. There is a perception that:

  - 8.4.4 measures "seat time"- the amount of time students have sat in a classroom- than what students have actually learned or failed to learn.

  -In 8.4.4 students' advances are based on their age, regardless of what they have learned.

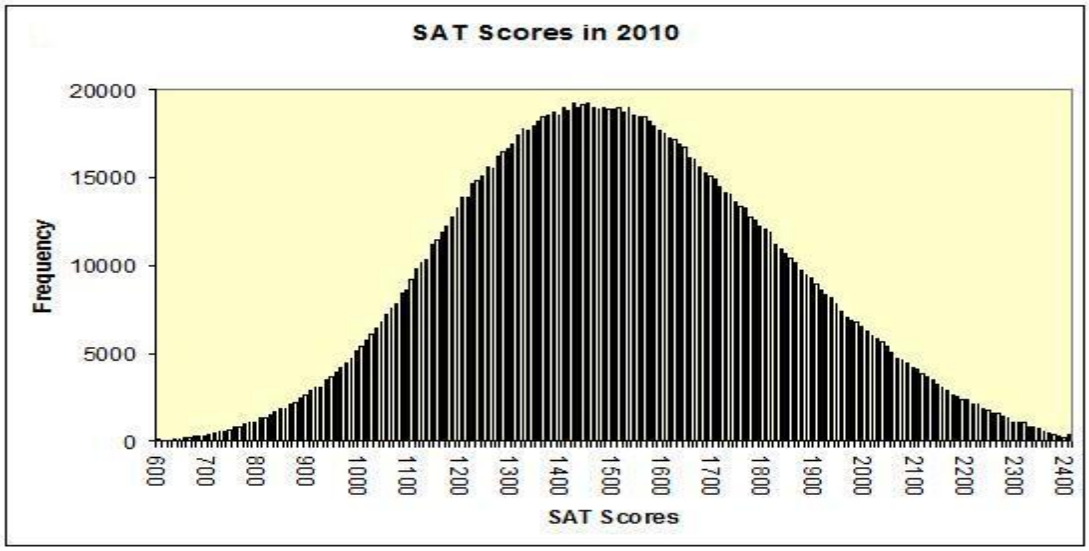
  NOTE: The proposed CBE is not tied to seat time.
- ***Item Banking***: Up to 2016 KNEC used item banking system. This an international practice. In 2016, CS directed KNEC to abolish item banking on the assumption that it is the source of examination leakages.

A number of these criticisms are misconceptions and ignorance of KNEC's strategies to achieve fair examinations.

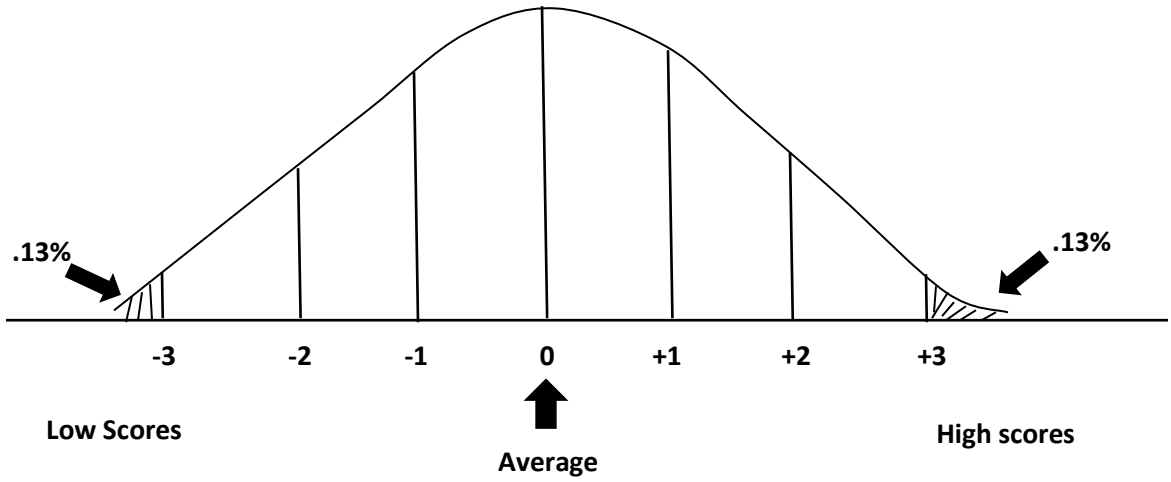
RECENT DEBATE ON KCSE PERFORMANCE

The Concept of a Normal Curve in Examination Results

In 19th century while studying individual differences, a psychologist by the name of Sir Francis Galton (1822-1911) came up with a concept of normal distribution of human behaviors. A normal curve is a statistical model that is used throughout the physical and social sciences. The graph below shows the scores of Scholastic Aptitude Test (SAT) in 2010. SAT is used by USA universities for selection/admission for undergraduate programmes.



The 2016, 2017 and 2018 KCSE examination results did not reflect a normal distribution. Abilities or behaviors of a large population assume a normal distribution, unless there are biases. When a distribution is normal, it should be like the bell-shaped figure given below.



In a normal distribution, the proportion of students below -3 standard deviations is the same as the number of students above +3 standard deviations.

KCSE PERFORMANCE TRENDS 2014-2019

Year	No. A	No. C^+
BEFORE MILITIRAZATION OF EXAMINATION ADMINISTRATION		
2014	3,073	149,717
2015	2,636	165,766
MILITIRAZATION ERA		
2016	141	88,929
2017	142	70,073
2018	315	90,377
2019	627	125,746

Source: Daily Nation, Thursday, December 19,2019, p.4.

2019 KCSE

2019 KCSE candidates=697,222

Overall national grade summary for 2019 KCSE Examination

sex	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E
F	269	2172	5145	9803	14961	21425	32084	43083	51813	69809	76198	12936
M	358	3624	8221									

A⁻ to A =6,423 =0.921

A-=5,796

C+= 125,746=18%

C- and below=541,476= 78.66%

Conclusion=Skewed

Gender:

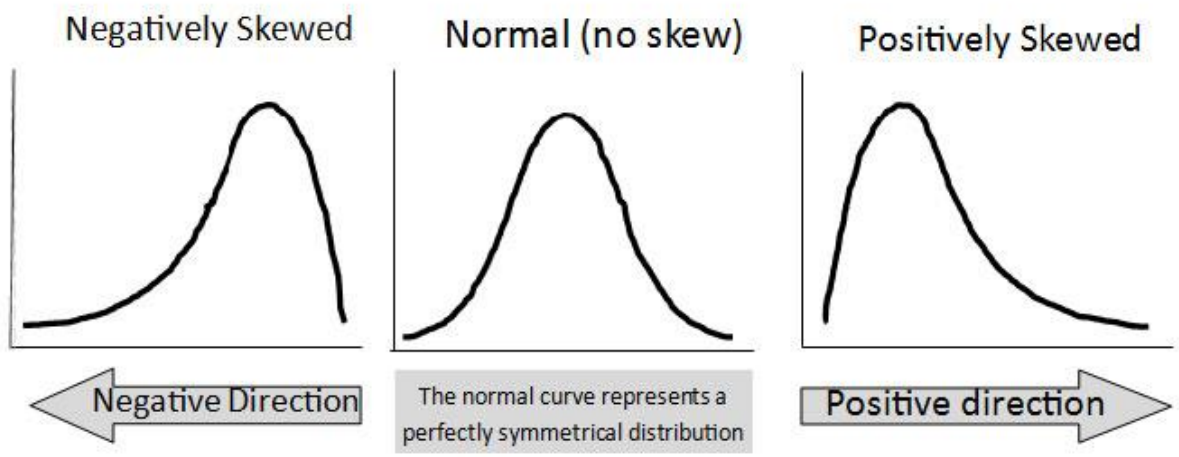
Boys=355782 (51%)

Girls-341,440 (49%)

2016 KCSE

The 2016 KCSE examination results reflected on extremely positively skewed distribution (**Source:** Economic Survey 2017, p. 51-52).

The extreme ends of the curve are not comparable – 141 (0.02%) vs 33,399 (5.85%). A and A-=4,786 (0.93%) and D⁻ and E= 183,328 (32.10%).



Causes of Non-Normality

1. ***Poorly set examination***. This is possible in 2016 examination where the setters copied test items from some commercial revision texts. Newspapers printed out the exact papers of the commercial texts. Hence those schools that used those books had unfair advantage. To hide this flaw no extra examination papers were left in schools. Because KNEC was directed to disregard ***Item Banking*** practice the examinations were developed in hurry.
2. ***Lack of content validity***. This means the examination items were drawn from outside the content taught.
3. ***Difficult examination*** – leading to the majority of students performing poorly. Difficulty examination does not discriminate bright students from dull students.
4. ***Poor marking conditions***. This is a possibility. Reports indicate that markers worked for long hours (6.00 am – 1.00 am) and experienced fatigue. Under these hostile conditions most makers are not willing to participate in 2017.
5. ***Poor coordination*** of examiners and hence failure to adhere to the marking scheme. Evidence gathered by KNUT shows that there was no standardization and moderation of the marking schemes.
6. ***The ignoring of award stage*** – where the chief examiners of every paper look at the performance in their subjects across the country and then propose the grading system. This is meant to normalize the grades. In 2016, this stage was ignored in a hurry to release the results for political mileage. It is not possible to use a grading system for Mathematics on English. Performance varies and therefore it must be moderated and standardized.

   Ignoring award ceremony/stage means that ***one grading system*** was applied and thus leading to erroneous conclusion that the poor performance is attributable to elimination of cheating.
7. Lack of score As in English for the first time in KCSE history is an indication of ***flawed marking***. English is not one of those subjects where cheating has been rampant in the past. So this cannot be attributed to past cheating.
8. ***KCSE is not criterion-referenced*** examination and hence predetermined grade cannot be used for mastery or non-mastery of skills. It is a norm-referenced examination. This is how KCSE has been treated in the past. It appears that it was treated as criterion test in 2016.

MASS FAILURE IN 2016, 2017 AND 2018 KCSE EXAMINATIONS

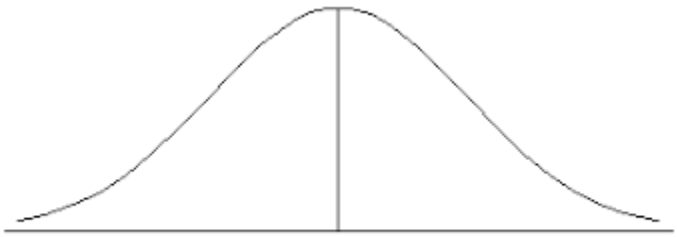
THE ABNORMALITY OF THE 2016, 2017 AND 2018 KCSE EXAMINATION RESULTS

The issues surrounding the debate and the politics of the 2016 and 2017 KCSE are:

1. Absence of a normal curve distribution of scores
2. Impact of examination cheating
3. Impact of absence of examination cheating
4. Norm-referenced testing
5. Criterion referenced testing
6. Non- standard practices

Normal Curve

In presenting the 2017 results to the President on December 20, 2017, the Education Cabinet Secretary was reported by the print media as having told him that the results were normally distributed. This statement is false. A normal curve must take a bell-shaped distribution as shown below.



mean = median = mode

In a normal distribution:

a) Proportion of those achieving A and A- and D- and E should have been equal – approximately 0.13 % on each side of the curve. This was not the case in 2017 KCSE:

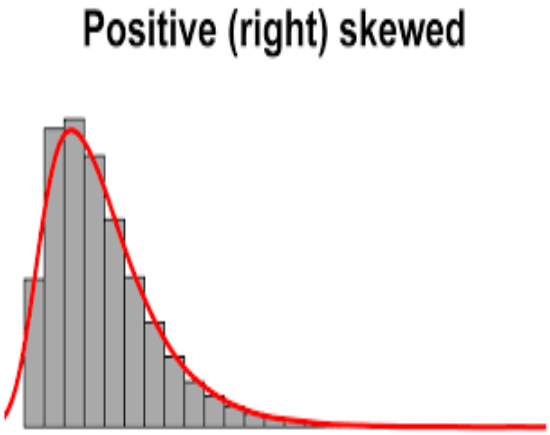
A and A-	=	0.5%
D- and E	=	35%

b) Assuming that students were well taught and well tested, they should get at least 50% of required knowledge and skills and that would translate to C

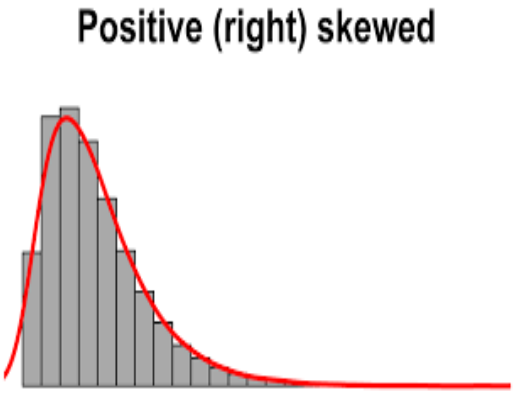
average. This was not the case in 2016 and 2017 KCSE, where the average was D+.

2016 KCSE Distribution and 2017 KCSE Distribution

a. 2016 KCSE



b. 2017 KCSE



Low Scores **High Scores**

Why Skewed Distribution?

The Cabinet Secretary for Education described KCSE skewed performance as the reality of the ability of the candidates. There are many factors that can bring out this trend including as discussed earlier in this course:

- Difficult questions
- Poor teaching
- Poor content validity
- Rushed marking for political expedience.
- Poor marking
- Use of commercial test items that give advantage to some schools. This was the case in 2016.
- Fatigued examiners
- Lack of standardization and moderation of test items
- Non-adherence to standard practices/procedures
- Treating KCSE as criterion-referenced test rather than as normed-referenced test.

Impact of Poor Grades Vs Future Opportunities

Labels that individuals achieve in examination are either enablers or disablers. In a society that suffers from what has been referred to as **"diploma disease"**, achievement of poor grades denies individuals future training opportunities. For 2016 and 2017, approximately 1 million four leavers achieved less than C^+ grade. These individuals have bleak future as their low grades deny them opportunities for admission in training institutions. This is one indicator of poor efficiency of an education system. This is a very high wastage rate in our education system. In the last two years, 1,062,000 candidates achieved D grade.

Where is the Future of this D Group?

The government is promising them opportunities in polytechnics and progression from certificates through diploma to degree level. This is unrealistic and unachievable promise:

i. Polytechnics do not have absorption capacity. In 2016, all polytechnics admitted only 28,000.
ii. Not all students have the interest and the aptitude for vocational and technical training.

iii. Progression from certificate level to degree is both expensive and time consuming.

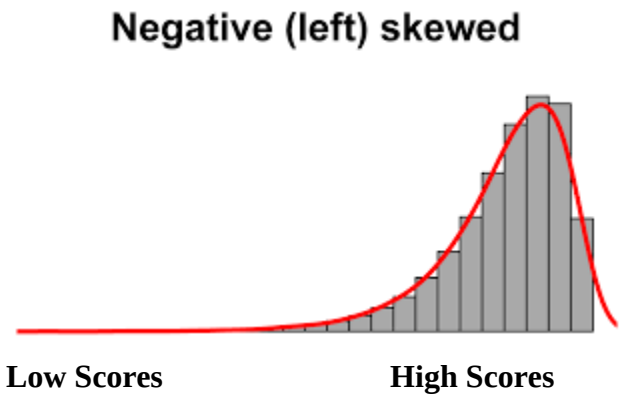
iv. Unrealistic to assume that opportunities for further education and training for these large numbers will be available in future.

v. Erratic policy changes on progression qualifications where earlier grades are used as the basis for admission for further education and training. Meaning those with poor grades at early stages will permanently be assigned to oblivion. Those who achieved four under KCE system of education have found it difficult to advance their education and training.

vi. The country has a poor record of non–recognition of experiential and prior learning for distance learners.

Impact of Examination Cheating

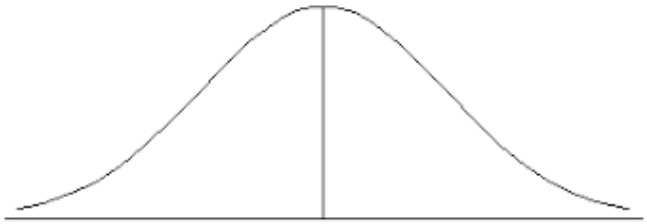
When cheating is a factor in an examination, the results ***cannot be normally distributed***. The results will be negatively skewed- a situation where more students will have higher scores.



Impact of Absence of Examination Cheating

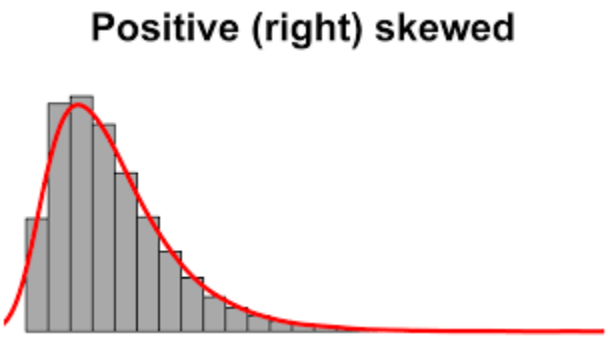
In 2016 and 2017 cheating was minimized to a non-significant level. In the absence of cheating, ***a true measurable behavior of learners should be normally distributed***. But this has not been reflected in 2016 and 2017 KCSE examinations. Instead, the results reflected positive skewness, a situation where more candidates achieved low scores- only 10% of the candidates (70,000 out of 651,000) had good grades for university. ***In non-cheating examination situation, the results***

should yield normal distribution, yet it is not the case in 2016 and 2017 KCSE. Is there something wrong with the whole assessment? Read Ollows article on "Is it the teaching and learning that is inadequate?" (Daily Nation, Sunday, December 24, 2017, p. 30). (See next page/below).



mean = median = mode

But for KCSE we have

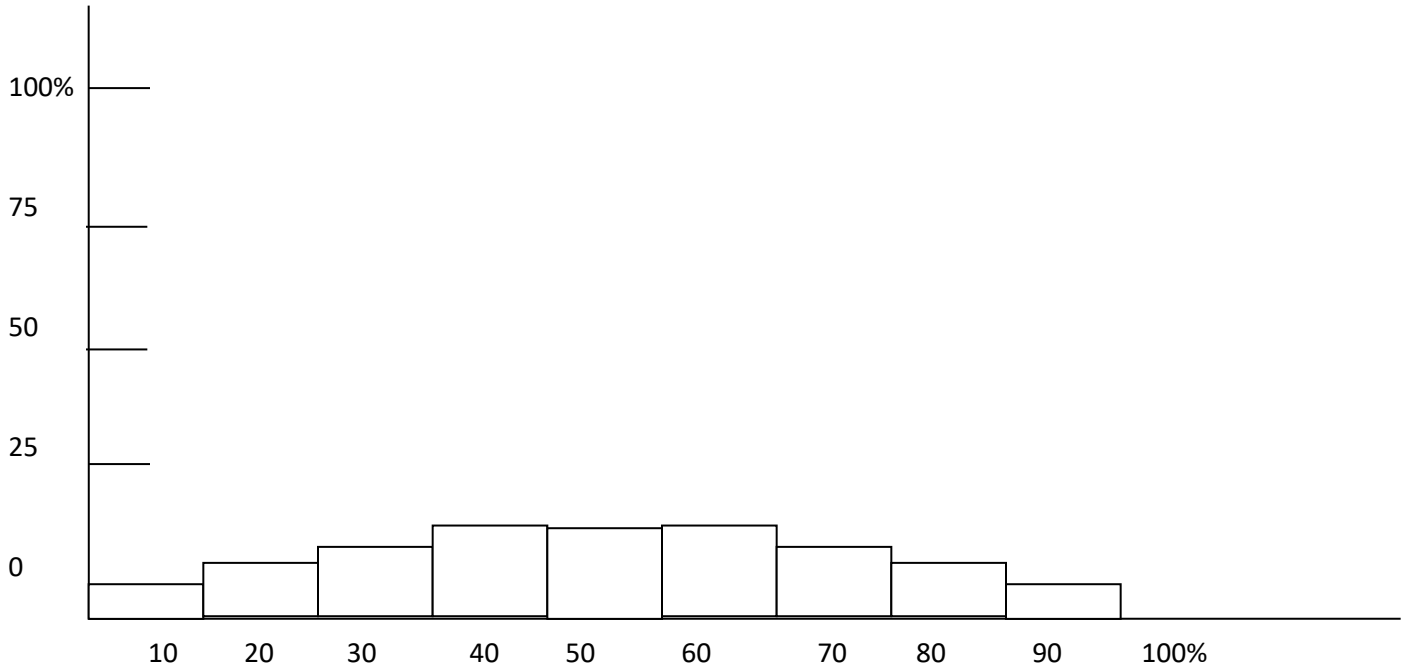


Low Scores **High Scores**

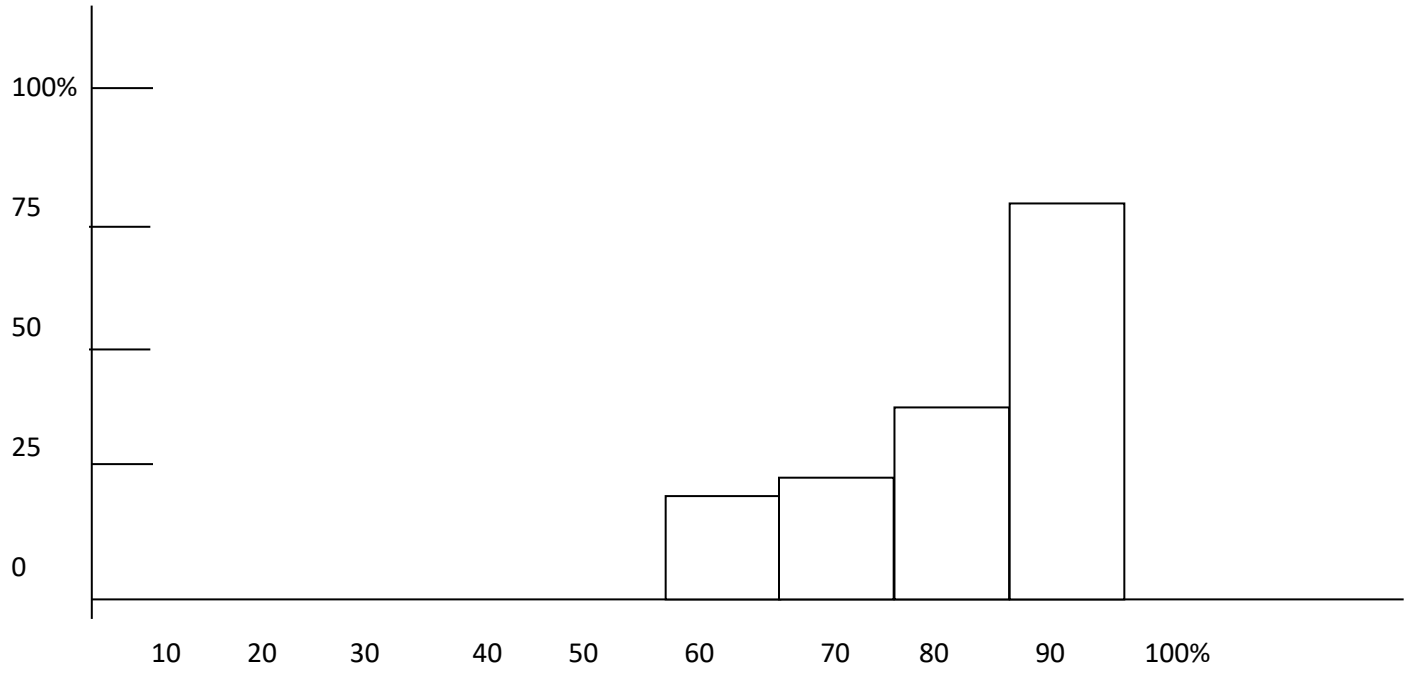
Treating KCSE as Criterion-Referenced Test rather than as Norm-Referenced Test

KCSE examination is a norm-referenced test and the results are expected to be evenly distributed according to the bell curve. The test aims at comparing the students with each other and not against specified /pre-set standards. There are no specified or declared standards for KCSE to qualify as criterion-referenced test. Read Ollows article given below.

(i) Norm-Referenced



Criterion-Referenced



The **criterion referenced assessor** would be reasonably pleased with the skewed distribution of the scores because his intention was to show whether pupils had learned well enough to attain the objectives. The **norm referenced assessor** prefers to set his test to ensure a wider spread of scores over a broader band of objectives (normal curve), such as is necessary for grading results in public examinations. He will be able to tell whether one student is either more or less expert than another, but cannot in the end say what aspect of the subject matter has been mastered or expertise acquired. In other words, he will not be able to say whether this or that objective has been attained.

UNANSWERED QUESTIONS ON 2017 KCSE RESULTS

Is it the teaching and learning that is inadequate?

FRED OLLOWS

The Cabinet Secretary for Education, Dr Fred Matiang'i, has just released the 2017 Kenya Certificate of Secondary Education (KCSE) examination results. This is about three weeks after the last paper.

This is a remarkable feat for the 20,000 markers of 610,501 candidates' papers – more than 8.5 million scripts. There are accolades about test security and marking efficiency. However, are the two the only factors for consideration in this whole exercise?

Testing and assessment is a core part of the education process. Linda Darling-Hammond, Professor at Stanford Graduate School of Education and President of the Learning Policy Institute, reckons that it is part of the accountability system in learning.

In its simplest form, testing is an indicator of how much learning has taken place. In education studies, it is a big time unit! There are principles that guide it. In turn, it provides an information mine. So, what do the KCSE test results tell us?

First, let me build context. KCSE is one of two general forms of testing – formative and summative. Formative is ongoing, continuous testing while summative is the testing at the end point of a learning cycle.

In the KCSE case, it is at the end of secondary school learning. Summative testing could well be at the end of a school year. Not surprisingly, it is compared to formative testing, a high stakes method.

High stakes issues tend to invite high stakes responses. In our Kenyan case, it is now commonly accepted that one of the responses has been for schools, parents and students to corruptly acquire the KCSE tests in order to beat the testing system's primary intentions.

The capacity to learn, best captured by cognitive (mental) intelligence, is normally distributed along uniform criteria like age. Therefore, a properly set test for any group of candidates will produce normal results. What does this mean?

That, if one plots the overall results of all the candidates, they should produce a normal curve. A normal curve looks like a bell which is why it is called a bell-shaped curve. It starts from an understanding that data will behave in a particular way around the average.

Now, Dr Matiang'I needs to tell us what form of testing the current KCSE is applying. Does the testing aim to compare the students with each other? In this type of testing, the students are compared with a representative sample of their peers – a "norm" group. Their results are expected to be evenly distributed according to the bell-curve. These are norm-referenced tests.

> "Education CS Dr Fred Matiang'i needs to explain which one of the big issues is problematic."

Alternatively, Dr Matiang'i's testing may be based on standards. That, by Fourth Form, a KCSE candidate should have acquired a set standard of knowledge and skills, in the various subject disciplines. The test results here meant to measure the mastery of the same according to the declared standard. Hence, standardised, or criterion-referenced testing.

Either way, there are serious questions hanging in the air. The average grade is just about a D+. If we use the common logic that an ordinary student, properly taught and tested, will get at least 50 pc, that would translate to a C aggregate. If the grades were norm-referenced, the number of grades A and A- should equal the number of grades D- and E.

What do we have instead?

A and A- together account for 0.5 pc (half of 1 pc) while D- and E account for 35 pc. This would be outrageous in normalised testing. If the KCSE aims for standardised testing, the D+ average means that the vast majority of the candidates have mastered much less than the 50 pc of the required knowledge and skills.

Thus, without regard to a host of other issues, the most contentious is that Dr Matiang'i, and the whole set up, can't answer, or aren't answering why their testing is producing abnormal results!

From these test scores, only about 10 pc of the candidates will have good enough grades for university. In a normalised environment, this would mean a proportion (6.5 pc) of above average candidates have been adjudged below university standard. How can that be?

It's clear that one, or more things, is wrong with this whole assessment. Test security is just one aspect of testing. As a matter of fact, properly, and ethically done, testing can even be an open book.

So, Dr Matiang'i needs to explain which one of the big issues is problematic: Is it the teaching (and learning) environment that is inadequate or, the testing that is not standard?

Mr Ollows, MSc (Organisational Behaviour) prepared the top KCSE class of 2001

NATION, SUNDAY 24/12/17
p. 30.

NON-STANDARD PRACTICES/PROCEDURES

Education > Markers allege they were made to work long and hard, compromising results

Teachers: Our pain in marking KCSE

The non-standard procedures adopted may have affected 2016 and 2017 KCSE examination results. ***Read*** Daily Nation, Saturday, December 23, 2017, p. 1 on the Teachers Pain in Marking KCSE. These non-standard procedures include *inter alia:*

1. **Fatigueness of the Examiners**
   i. ***Standard procedure:*** Examiners as human beings require time for breaks and relaxation. In the past, marking would start at 8.00a.m and end between 4.00p.m. and 7.00p.m.

   ii. ***Abnormal procedure:*** It was reported that "examiners who handled English Papers 3, one of the most taxing examinations, worked from 4.00 a.m. to 10.00 p.m. (16 hours each day).

      a) This lead to wide-spread fatigue. Fatigueness allows mistakes to pass and wrongly awarding of marks.
      b) Examiners who required medical attention were given within the marking centre. The death of one examiner was attributed to marking pressure (Daily Nation, Saturday, December 23, 2017, p.4-5.

2. **Lack of established marking standards or uniformity**
   i. ***Standard procedure:*** Traditionally, the examiners are required to take two days to familiarize themselves with the marking scheme by going

through dummies. Marking scheme must be congruent with the questions.

ii. ***Abnormal Procedure:*** In 2017 KCSE the familiarization only took a half a day. Therefore inconsistencies in marking were bound to happen.

3. **Mismatch between the some test items (questions) and expected responses (answers)**

   i. ***Standard Procedure:*** Validity and reliability can only be achieved if the question set and the answer match. One examiner reported that in 2017 KCSE there were errors in the marking schemes but these could not be corrected because there was no time to do it. ***The emphasis appears to have been on the markers to finish marking and have the results released in record time*** (Daily Nation, Saturday, December 23, 2017, p.4)

   ii. ***Abnormal procedures*****:** It was reported that "at Moi Girls School in Nairobi, where History Paper 2 was seeing marked, examiners realized that one question had the wrong answer. However, nobody cared to do the correction as the push was to ***finish marking as quickly as possible***. The implication is that candidates were disadvantaged. Moderation of marking would have taken case of this error-through compensation. The Ministry has removed this key process. The Cabinet Secretary (CS), unfortunately called moderation process "massaging" of results.

4. **Lack of Inter-examiners' Reliability Process**

   i. ***Standard Procedure:***

   During marking, examiners are put in a pool of seven with a team leader. For every 10 scripts marked, the team leader has to review (re-mark) at least two scripts which are picked randomly to verify if they have been marked well. The marking errors allowed is – 2 or +2 and if an examiner goes

beyond this margin, he/she is forced to re-mark the entire scripts. If she/he consistently makes the same mistake he/she is retired (expelled) from marking exercise.

ii. ***Abnormal Procedure:***

In 2017 KCSE, it was reported by the examiners that the inter-examiners' process was not done. Skipping this process enabled many examiners go for volume of scripts since payment is based on the number of scripts marked.

COMPETENCE BASED EDUCATION/CURRICULUM (CBE)

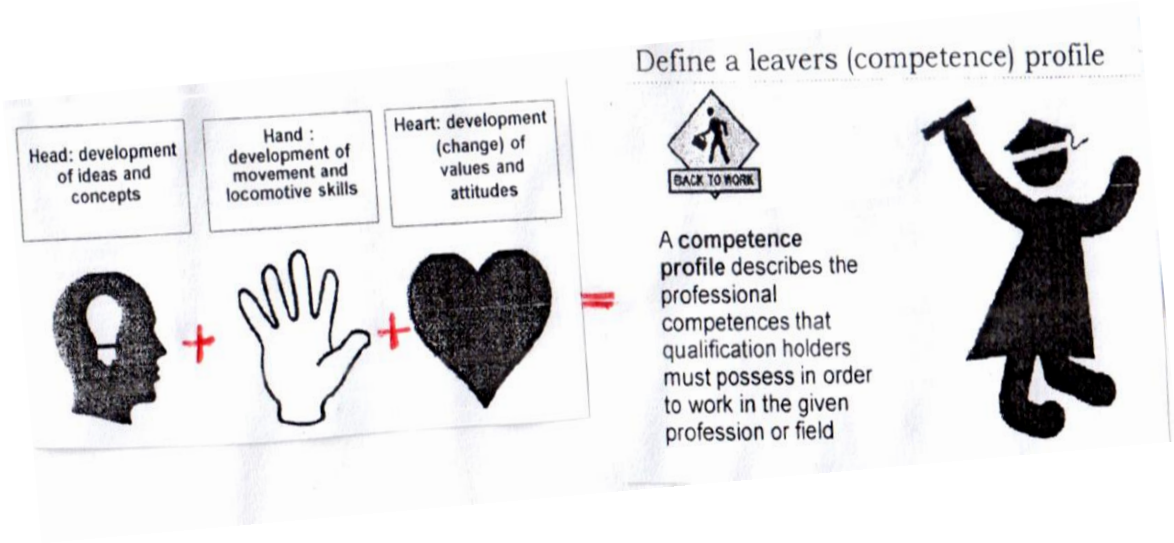
Kenya plans to phase out 8-4-4 and introduce CBE. Why have the previous reforms in Kenya failed?

- We abolished 'A level' segment saying it did NOT serve Kenya well. 'A level' serves Britain well and enables Britain to advance technologically. So why is it not good for Kenya? Why is it that it develops Britain and under develops Kenya?
- We changed to 8-4-4. It is an education structure that serves USA and Canada well? Why is it that it develops USA and Canada and under develops Kenya? Education reforms in Kenya are guided by politics rather professionalism.

What is Competence–Based Competence (CBC) and what makes it different?

The most important characteristic of competency-based education is that it ***measures learning rather than time***. Students' progress by demonstrating their competence, which means they prove that they have mastered the ***knowledge, skills and attitudes (called competencies)*** required for a particular course, regardless of how long it takes. While more ***traditional models can and often do measure competency, they are time-based*** — courses last about eight years, and students may advance only after they have put in the ***seat time***. This is true even if they could have completed the coursework and passed the final exam in half the time. So, while most colleges and universities hold time requirements constant and let learning vary, competency-based learning allows us to hold learning constant and let time vary.

Head + Hand + Heart= Competent Professional



Competence= combination of knowledge, skills and attitude required to become a competent professional. [Refer Nelson Mandela on mother tongue].

Benefits of Competency-Based Learning

There are many reasons schools are moving away from seat time and toward competency-based learning. These reasons include:

- This is the type of education that combines the above in a graduate.
- Performance based/outcome-based education with clear performance indicators.
- Education that incorporates real life assignments and assessment (problem solving skills, meaningful, etc).
- Education that incorporates the requirements of the labour market and industry.
- Education that gives students to learn at their own pace.
- Learner-centred education.
- Competency-based learning keeps each student from getting bored.
- Competency-based learning allows each student to work at his or her own pace.
- A greater mastery of the subject, done in a less frustrating and more invigorating way. Students move at their own pace with the curriculum.

In summary, it is argued by the proponents of CBE that Competency-based education (CBE) is a new model in education that uses learning, not time, as the metric of student success. This student-centered, accelerated approach redefines traditional credit-based requirements in learning and stresses competencies derived from the skills proven to be the most relevant by educators and employers.

With competency-based education, institutions can help students' complete credentials in less time, at lower cost — with a focus on real-world learning that leads to greater employability. This versatile model benefits the student, the instructor, the institution, and the economy. Very thing sounds good with CBE **but……**

Where has CBC succeeded?

CBE has been successfully implemented in South Korea, Japan, Finland, and Netherland. This is primarily because of the culture of the people. In South Korea children spend up to 14 hours in school (8.00 am to 11.00pm) per day. Since most children do not get home until midnight, dinner is served in school? Why? to get into good college.

South Korea Education Structure

1. Pre- schools (3-6 years old children= 2 years –(optional)
2. Primary Education = 6 years
3. Lower Esc (Middle School) =3 years
4. Upper Sec (High School) =3 years:

Academic Stream (62%)

Vocatioal Stream (38%)

Where has CBC failed?

CBE has had little success, if any, in Africa. South Africa and Malawi tried and abandoned. It was tried for 12 years in South Africa. Why it failed in South Africa?

- It involves too much administration and record keeping.
- Very expensive.
- Exodus of teachers. Many teachers left the teaching profession.

- For 12 years' teachers never understood how to implement CBE.
- Assessment very subjective.

Challenges of CBC

1. Failed system

It is also important to note that the proposed Competence-Based Curriculum education was tried for 12 years in South Africa and it failed/abandoned. In South Africa it was called Objective-Based Education (OBE). It was also tried in Malawi and abandoned.

The proposed Kenyan CBE was borrowed from Japan and South Korea. Kenya does not have the culture of these countries and is wrong to assume that we will succeed. Do we have the Asian culture that has been responsible for the success of CBE? No.

2. Assessment of student in CBC is subjective

In Kenya **70**% of assessment will come from CA. The proposed competence-based curriculum education system proposes the replacement of national examinations as a tool for testifying the achievement of an educational level with continuous assessment. As a country we are at a stage where the government does not respect and trust teachers. The invigilation and the marking of the 2016 KCSE examinations is a clear demonstration of this mistrust. In pointing out his level of mistrust of teachers the CS in charge of Education said, "**With the use of ICT, we eliminated people who were changing marks"** (Standard, January 12, 2017: p.8). The people being referred to are teachers. These are same people that the proposed CA system will rely on. Will ICTs be able check CAs awards?

Is this CA policy direction not paradoxical? Will the teachers' assessment be a fair tool for assessment in a country with limited resources? Will this approach eliminate competition in the job market place? What about corruption? What about the "halo effect"?

The use of CA works well in countries where recruitment is not strictly based on certificates but on what a candidate can do. This approach of CA failed when it was used by KNEC in Primary Teachers Examinations in early 1980s. At this time 40% of the final grade came from CA and 60% from KNEC examination. Teacher trainees were passing CA but failing the KNEC examinations.

3. **Low public participation**. This proposed system is being driven politically. No sessional paper was produced before rolling it out. This ie expected in June 2019- three years after its implementation.
4. **Too futuristic** and impractical to implement in Kenya in its current form.
5. **Expensive to implement.**
6. **Overloaded syllabus**. Between 11-12 subjects to be taken.
7. **Pupils' progression** from primary to secondary unclear.
8. Teacher-based **subjective assessment** in determining movement from primary to secondary is bias and hence pupils cannot go to their school of choice in the absence of a standard, unifying national examination.
9. **Damage national integration** because children will remain in their neighborhoods.
10. **Demand literacy of parents.**

https://www.youtube.com/watch?v=Da_f-wApT9c

https://www.youtube.com/watch?v=DOTBxnBxPL0

review questions

a. Discuss the current place of national examinations in Kenya.
b. Discuss public criticisms of national examinations in Kenya.
c. Explain non-normality in examinations.
d. Discuss the proposed Competency Based Education.

Further Reading

1.Education Measurement and Evaluation : J.Swarupa Rani, Discovery Publishing house.

2. Measurement and Evaluation in Teaching : Norman Edward Gronlund Macmillian

3. Measurement and Assessment in Teaching : Robert L. Linn Pearson Education India.

4. Program Evaluation and performance measurement : James C. Me. David, Laura

R.L. Hawthorn, Sage Publication